



Seminar

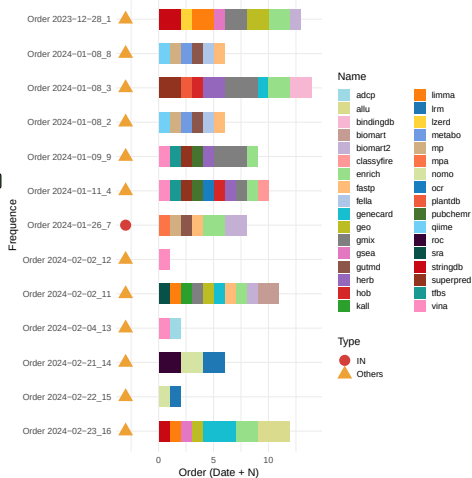
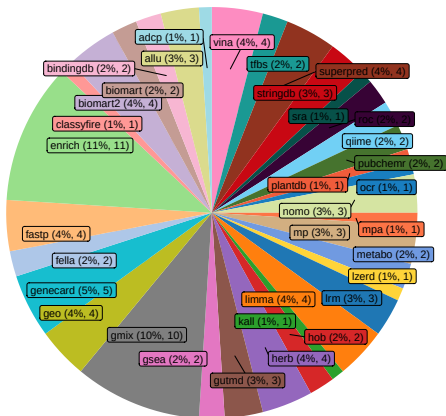
Lichuang Huang

Wie-Biotech

2024-03-13

Tasks summary

Overview 2024-01 ~ 2024-02



New Methods

- gmix: DisGeNet, GeneCards, PharmGKB
- lzerd: LZerD and HawkDock webserver, protein–protein docking
- hob: Python, human oral bioavailability
- plantdb: (crawler) The Database PlantaeDB
- superpred: (crawler) drug-targets relationship prediction (SwissTargetPrediction)
- ocr: Python, doctr
- tfbs: (crawler) Transcription Factor Target Gene Database

New Methods

- mpa: MetaPhlAn, microbial communities from metagenomic data
- adcp: auto molecular docking (docking peptides)
- nomo: logistic, nomogram
- classfire: systematic classification of chemical compounds
- msconvert: file formats of MASS data
- xcms: MASS data preprocessing

Article sharing

Reconstruction of colon cancer microbiome

RESEARCH

Open Access

Location and condition based reconstruction of colon cancer microbiome from human RNA sequencing data



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Abstract

Background The association between microbes and cancer has been reported repeatedly; however, it is not clear if molecular tumour properties are connected to specific microbial colonisation patterns. This is due mainly to the current technical and analytical strategy limitations to characterise tumour-associated bacteria.

Methods Here, we propose an approach to detect bacterial signals in human RNA sequencing data and associate them with the clinical and molecular properties of the tumours. The method was tested on public datasets from The Cancer Genome Atlas, and its accuracy was assessed on a new cohort of colorectal cancer patients.

Results Our analysis shows that intratumoural microbiome composition is correlated with survival, anatomic location, microsatellite instability, consensus molecular subtype and immune cell infiltration in colon tumours. In particular, we find *Faecalibacterium prausnitzii*, *Coprococcus comes*, *Bacteroides* spp., *Fusobacterium* spp. and *Clostridium* spp. to be strongly associated with tumour properties.

Conclusions We implemented an approach to concurrently analyse clinical and molecular properties of the tumour as well as the composition of the associated microbiome. Our results may improve patient stratification and pave the path for mechanistic studies on microbiota-tumour crosstalk.

Keywords Tumour microbiome, RNA-Seq data deconvolution, Microbe-tumour interaction, Microbiome biomarker

Details

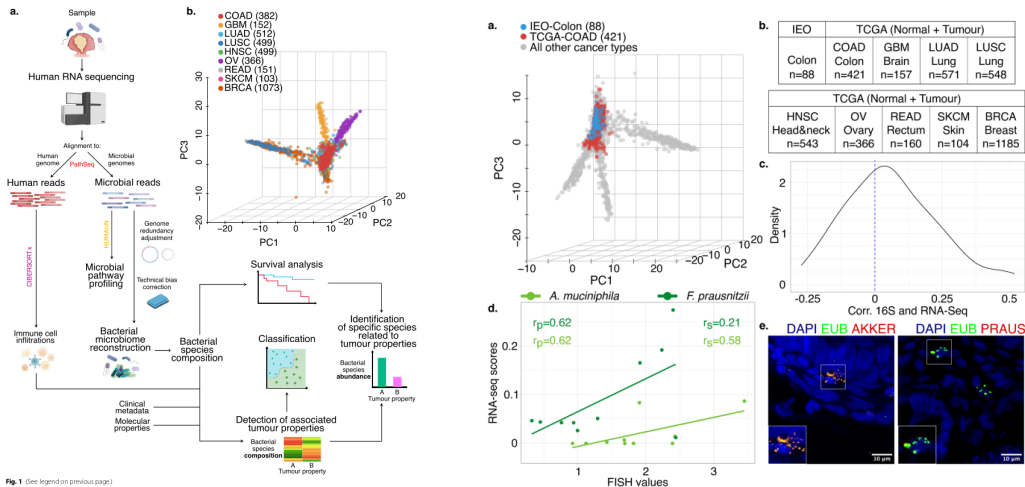


Fig. 1 (See legend on previous page.)

Code availability

  SamGa3 / microbiome_reconstruction

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 **microbiome_reconstruction** Public

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fc9fd2 · 3 weeks ago

296 Commits

conda	add conda yml multiplatform	2 years ago
data	Delete data/RNAseq/TPM/tcga directory	4 months ago
metadata	upload files	3 weeks ago
scripts	Update paths	4 months ago
.gitignore	modified gitignore	3 weeks ago
LICENSE	Initial commit	2 years ago
README.md	Update README.md	3 weeks ago
requirements.txt	add requirements file	2 years ago

[README](#) [GPL-3.0 license](#)

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Schedule

Schedule

- Finish orders
- New methods

Thanks